2. Find the LCM of the given expressions.

(i) $4x^2y,\ 8x^3y^2$ (ii) $9a^3b^2,\ 12a^2b^2c$ (iii) $16m,\ 12m^2n^2,\ 8n^2$

(iv) $p^2-3p+2,\ p^2-4$ (v) $2x^2-5x-3,\ 4x^2-36$

(vi) $(2x^2-3xy)^2, (4x-6y)^3, 8x^3-27y^3$

3.3.3 Relationship between LCM and GCD

Let us consider two numbers 12 and 18.

We observe that, LCM $(12,18)=36$, GCD $(12,18)=6$.

Now, LCM $(12,18)\times$ GCD$(12,18)$ $=36\times6=216=12\times18$

Thus LCM × GCD is equal to the product of two given numbers.

Similarly, the product of two polynomials is the product of their LCM and GCD, That is, $f(x)\times g(x)=LCM[f(x),g(x)]\times GCD[f(x),g(x)]$

Illustration

Consider $f(x)=12(x^2-y^2)$ and $g(x)=8(x^3-y^3)$

Now $f(x)=12(x^2-y^2)=2^2\times3\times(x+y)(x-y)$ ...(1)

and $g(x)=8(x^3-y^3)=2^3\times(x-y)(x^2+xy+y^2)$ ...(2)

From (1) and (2) we get,

$$\begin{aligned}\text{LCM}[f(x),\ g(x)] &= 2^3\times3\times(x+y)(x-y)(x^2+xy+y^2)\\ &= 24\times(x^2-y^2)(x^2+xy+y^2)\\ GCD\left[f(x),g(x)\right] &= 2^2\times(x-y)=4(x-y)\\ LCM\times GCD &= 24\times4\times(x^2-y^2)\times(x^2+xy+y^2)\times(x-y)\\ LCM\times GCD &= 96(x^3-y^3)(x^2-y^2) \quad ...(3)\end{aligned}$$

product of $f(x)$ and $g(x)=12(x^2-y^2)\times8(x^3-y^3)$

$$=96(x^2-y^2)(x^3-y^3) \quad ...(4)$$

From (3) and (4) we obtain LCM × GCD $=f(x)\times g(x)$

Thinking Corner

Is $f(x)\times g(x)\times r(x)=$LCM $[f(x),g(x),r(x)]\times$ GCD $[f(x),g(x),r(x)]$?

Exercise 3.3

1. Find the LCM and GCD for the following and verify that $f(x)\times g(x)=LCM\times GCD$

(i) $21x^2y,\ 35xy^2$ (ii) $(x^3-1)(x+1),\ (x^3+1)$ (iii) $(x^2y+xy^2),\ (x^2+xy)$

2. Find the LCM of each pair of the following polynomials

(i) $a^2+4a-12,\ a^2-5a+6$ whose GCD *is* $a-2$

(ii) $x^4-27a^3x, (x-3a)^2$ whose GCD is $(x-3a)$

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